

Assignment 4

Thursday, April 28, 2023 4:12 AM

3.77 From heat evolution in x -direction we have

$$\frac{d^2 T}{dx^2} + \frac{\dot{q}}{k} = 0$$

where $\dot{q}$ = volumetric heat

$$\frac{d^2 T}{dx^2} = -\frac{\dot{q}}{k}$$

$$\frac{dT}{dx} = -\frac{\dot{q}_0}{k} \left(1 - \frac{x}{L}\right)$$

integrating both sides once

$$\frac{dT}{dx} = -\frac{\dot{q}_0}{k} \left(x - \frac{x^2}{2L}\right) + C_1$$

again

$$T = -\frac{\dot{q}_0}{k} \left(\frac{x^2}{2} - \frac{x^3}{6L}\right) + C_1 x + C_2$$

Boundary conditions

$$x = L, \frac{dT}{dx} = 0 \text{ (insulated)}$$

$$\therefore C_1 = \frac{\dot{q}_0 L}{2k}$$

$$\text{at } x = 0, T = T_0 \Rightarrow C_2 = T_0$$

$$T(x) = -\frac{\dot{q}_0}{k} \left(\frac{x^2}{2} - \frac{x^3}{6L}\right) + \frac{\dot{q}_0 L}{2k} x + T_0$$

$$T(x) - T_0 = -\frac{\dot{q}_0}{k} \left(\frac{x^2}{2} - \frac{x^3}{6L}\right) + \frac{\dot{q}_0 L}{2k} x$$

3.80

Consider T at any point:

$$\frac{1}{x} \frac{d}{dx} \left(x \frac{dT}{dx} \right) = -\frac{\dot{q}}{k}$$

$$x \frac{dT}{dx} = -\frac{\dot{q} x^2}{2k} + C_1$$

$$\frac{dT}{dx} = -\frac{\dot{q} x}{2k} + \frac{C_1}{x}$$

$$\text{We have } \frac{dT}{dx} = 0 \text{ at } x = x_1$$

$$\therefore C_1 = \frac{\dot{q} x_1^2}{2k}$$

using this

$$\frac{dT}{dx} = -\frac{\dot{q} x}{2k} + \frac{\dot{q} x_1^2}{2k x}$$

integrating

$$T = -\frac{\dot{q} x^2}{2k} + \frac{\dot{q} x_1^2}{2k x} \ln x + C_2 \quad \text{--- ①}$$

$$\text{at } x = x_2$$

$$k \frac{dT}{dx} = h(T_w - T_\infty)$$

$\therefore$ we get

$$k \frac{dT}{dx} = -\frac{\dot{q} x}{2} + \frac{\dot{q} x_1^2}{2x} = h(T_w - T_\infty)$$

$$\frac{\dot{q}}{2k} \left(x^2 - \frac{\dot{q} x_1^2}{x_2} \right) = (T_w - T_\infty)$$

Substituting ① here

$$T_{x=x_2} = T_w = -\frac{\dot{q} x_2^2}{2k} + \frac{\dot{q} x_1^2}{2k} \ln x_2 + C_2$$

$$\therefore C_2 = \frac{\dot{q}}{2k} \left(x_2^2 + \dot{q} x_1^2 \ln x_2 \right) \Rightarrow \text{②}$$

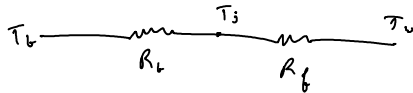
$$T = -\frac{\dot{q} x^2}{2k} + \frac{\dot{q} x_1^2}{2k} \ln x + \frac{\dot{q}}{2k} \left(x_2^2 + \dot{q} x_1^2 \ln x_2 \right)$$

$$\textcircled{b} \quad T = \frac{-q}{2k} (x_1^2 - x_1^2 \ln x) + \frac{q}{2k} (x_2^2 + q x_1^2 \ln x_2)$$

$$\frac{q}{2k} \left(x_1^2 - \frac{q x_1^2}{x_2} \right) = (T_w - T_\infty)$$

4.30

for case (a) :-



R_b - Base resistance

$R_f \rightarrow$ fin resistance

$$q_b = \frac{T_b - T_\infty}{R_{total}} = \frac{T_b - T_\infty}{R_b + R_f}$$

$$q_b = \frac{T_j - T_\infty}{R_f}$$

$$T_j = q_b R_f + T_\infty$$

For resistance 2D

$$R_b = \frac{1}{s k_c}$$

$$\Rightarrow s = 2D$$

$$(R_b)_{Al} = \frac{1}{2 \times 0.005 \times 240} = 0.4167 \text{ K/W}$$

$$(R_b)_{ss} = \frac{1}{2 \times 0.005 \times 15} = 6.67 \text{ K/W}$$

$$R_f = \left(\sqrt{h_f \lambda A_c} \right)^{-1}$$

$$\Rightarrow A_c = \frac{\pi}{4} (0.005)^2 = 0.19349 \times 10^{-9} \text{ m}^2$$

$$\Rightarrow h = h_{al} = 240 \text{ W/m}^2\text{-K}$$

$$\Rightarrow h = 55 \text{ W/m}^2\text{-K}$$

$$\Rightarrow p = \pi D = \pi \times 0.005 = 0.01571 \text{ m}$$

$$R_f = \left[\sqrt{55 \times 0.0157 \times 240 \times 0.19349 \times 10^{-9}} \right]^{-1}$$

$$R_f = 15.6715 \text{ K/W}$$

Case (b) (c) Aluminium

$$(q_b)_{Al} = \frac{T_b - T_\infty}{(R_b)_{Al} + R_f} = \frac{100 - 25}{0.4167 + 15.6715} = 4.68 \text{ W}$$

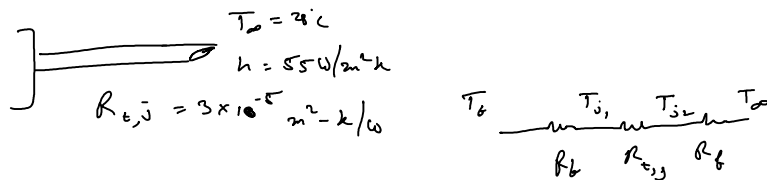
$$(T_j)_{Al} = q_{rad} R_f + T_\infty = 98.03^\circ\text{C}$$

(ii) Stainless Steel

$$q_{f,s} = \frac{T_b - T_\infty}{R_{fs} + R_f} = \frac{100 - 25}{6.67 + 15.6715} = 3.356 \text{ W}$$

$$\begin{aligned} T_{j,s} &= q_{f,s} R_f + T_\infty \\ &= 77.59^\circ\text{C} \end{aligned}$$

Case (b) :-



$$R_{t,j} = \frac{3 \times 10^{-5}}{0.136349 \times 10^{-4}} = 1.528 \text{ K/W}$$

$$\begin{aligned} q_f &= \frac{T_b - T_\infty}{R_b - R_{t,j} + R_f}, \quad T_{j,i} = q_f (R_{t,j} + R_f) + T_\infty \\ T_{j,e} &= q_f (R_f) + T_\infty \end{aligned}$$

Case b ① Aluminium

$$\begin{aligned} q_{f,Al} &= \frac{T_b - T_\infty}{R_{b,Al} + R_{t,j} + R_f} = \frac{100 - 25}{0.4167 + 1.528 + 15.6715} \\ &= 4.257 \text{ W} \end{aligned}$$

$$\begin{aligned} T_{j,Al,e} &= 4.257 \times 15.6715 + 25 \\ &= 91.713^\circ\text{C} \end{aligned}$$

$$\begin{aligned} T_{j,Al,i} &= 4.257 (15.6715 + 1.528) + 25 \\ &= 98.28^\circ\text{C} \end{aligned}$$

② Stainless Steel

$$\begin{aligned} q_{f,s} &= \frac{100 - 25}{6.67 + 1.528 + 15.6715} \\ &= 3.14 \text{ W} \end{aligned}$$

$$\begin{aligned} T_{j,s,e} &= q_{f,s} R_f + T_\infty \\ &= 3.14 \times 15.6715 + 25 \\ &= 74.2^\circ\text{C} \end{aligned}$$

$$T_{j,s,i} = 3.14 (15.6715 + 1.528) + 25$$

$$= 7.9^{\circ}\text{C}$$

